Additional file 2: TRIPOD+AI reporting checklist

Blood metal and nutritional biomarkers did not improve identification of low cognitive performance beyond demographics in older US adults: a pre-registered analysis of NHANES 2011-2014

Shreyan Kancharla, Neil Patel

TRIPOD+AI (Collins et al., BMJ 2024) is the reporting standard for prediction-model studies developed with machine learning, extending TRIPOD (2015). The "Where addressed" column points to the section of the manuscript, or the repository artifact, that satisfies each item; the analysis code and data referenced throughout are public at `https://github.com/NeilP211/nhanes-ieee`.

#	Item	Where addressed	Status
Title and abstract			
1	Identify the study as developing a prediction model, specify the target population and outcome	Title: "Reported biomarker-cognition associations do not translate into predictive utility: a pre-registered analysis of NHANES 2011-2014"	done
2	Structured abstract: objectives, data, participants, outcome, predictors, analysis, results, conclusions	Abstract (Background/Methods/Results/Conclusions, under DPR's 350-word limit)	done
Introduction			
3a	Rationale, including the clinical or scientific gap	Background section; seven studies report associations in this population, none isolating the biomarkers' incremental contribution over demographics; one 2025 study (Ren et al.) reports a combined-model AUC on a related population without that decomposition, addressed directly	done
3b	Objectives, including whether development, validation, or both	Development with internal validation by repeated cross-validation. No external validation set exists	done
Methods: data			
4a	Data source and rationale	NHANES 2011-2012 and 2013-2014, public	done
4b	Dates of data collection, and whether data were split	Two survey cycles, pooled. No train/test split; repeated 10-fold CV instead	done
5a	Eligibility criteria	Aged 60+, MEC examined (`RIDSTATR == 2`), at least 3 of 4 cognitive scores	done
5b	Setting and recruitment	NHANES complex multistage probability sample of the non-institutionalised US population	done
5c	Details of treatments received, if relevant	Not applicable. Observational, cross-sectional, no intervention	done
Methods: outcome			
6a	Outcome definition and how it was measured	Composite z-score of CERAD immediate, CERAD delayed, animal fluency, DSST; lowest quartile. An epidemiologic classification relative to the sample, not demographically normed (Methods, Outcome). Additional file 1, section 2	done
6b	Whether outcome assessors were blinded to predictors	Not applicable. Outcome measured by standardised test administration before any biomarker assay was linked	done
6c	Any actions to blind assessment of the outcome	Not applicable, as above	done
Methods: predictors			
7a	Predictors, including how and when measured	Methods, Predictors subsection. 25 features in three blocks	done
7b	Whether predictor assessors were blinded to the outcome	Laboratory assays performed by CDC contract laboratories with no access to cognitive results	done
Methods: sample size and missing data			
8	How sample size was arrived at	Not a power calculation: the analytic sample is the whole eligible population. $n = 1{,}784$ with 431 events, 17.2 events per feature	done
9	How missing data were handled	Complete-case on a single locked row set as primary; multiple imputation by chained equations fitted inside each fold as pre-specified sensitivity; complete cognitive battery analysis added at peer review. Missing cognitive scores by test, reason and group in Additional file 4 (Tables S2 to S5). Additional file 1, section 3	done
Methods: analytical methods			
10a	How predictors were handled in the analysis	Log transforms for right-skewed analytes, one-hot for categoricals, standardisation. All fitted inside folds	done
10b	Model type, building procedure, and internal validation	L2-penalized logistic (primary), random forest, XGBoost. Nested CV: inner CV for hyperparameters, outer 10-fold $\times$ 5 repeats	done

(continued)

#	Item	Where addressed	Status
10c	How any model updating or recalibration was done	None performed. Calibration reported as observed, not corrected	done
11	Measures used to assess performance	AUC, AUPRC, Brier, calibration slope and intercept, decision-curve net benefit	done
12	How heterogeneity or subgroups were handled	Two survey cycles pooled with cycle recorded; English-only administration as a pre-specified sensitivity analysis	done
Methods: specific to AI and machine learning			
A1	Rationale for the chosen algorithms	Penalized regression as the pre-specified primary, two tree ensembles to test whether flexibility helps at this sample size. Additional file 1, section 5	done
A2	Hyperparameter tuning: search space and selection procedure	Grids in the analysis code; selected by inner 3-fold CV on training folds only	done
A3	How data leakage was prevented	Every transform inside sklearn `Pipeline` objects fitted per fold. An automated audit refits a fold and asserts the scaler's fitted mean equals the training-fold mean and not the full-sample mean; it fails the run rather than warning, and it passed	done
A4	Computational resources and reproducibility	Single machine, no GPU, full pipeline under one hour. Master seed 20260831; versions pinned	done
A5	Model interpretability methods and their limitations	SHAP on the best tree model, with rankings bootstrapped over 200 resamples. The instability of biomarker ranks is reported as a finding, and the limitation of single-run SHAP is stated explicitly (Fig. 5)	done
A6	Fairness considerations across subgroups	Race and ethnicity included as a predictor; the eGFR equation used is CKD-EPI 2021, which deliberately omits the race coefficient. No subgroup fairness audit was pre-specified. Declared as a limitation in the Discussion	done
Open science			
13a	Funding and role of funders	Declarations: unfunded	done
13b	Conflicts of interest	Declarations: none declared	done
13c	Protocol and registration	Additional file 1, dated 2026-08-31 before any outcome-predictor relationship was examined, with an explicit integrity statement to that effect	done
13d	Data availability	NHANES is fully public; the download script reconstructs the raw data from CDC URLs	done
13e	Code availability	Complete pipeline, independently runnable, public on GitHub	done
Results			
14a	Participant flow, ideally a diagram	Fig. 1	done
14b	Participant characteristics, including missingness	Table 1 (by outcome status, including the cognitive phenotype); Table 2 (included vs. excluded); Additional file 4 (Tables S2 to S5)	done
15	Number of participants and outcome events	1,784 participants, 431 events, prevalence 0.2416	done
16	Model specification: all coefficients or a means to obtain them	Additional file 3 (Table S1): full M3 logistic regression refit on the complete analytic sample, standardised coefficients, odds ratios, 2,000-resample bootstrap 95% CIs	done
17	Model performance with uncertainty	Results (Discrimination, Margin analysis, Algorithm comparison); Table 3; Figs. 2 to 4; Additional file 4 (Tables S6 to S8)	done
18	Results of any model updating	None performed	done
Discussion			
19	Limitations	Discussion: cross-sectional design; complete-case selection is mildly non-random; age top-coded at 80; no external validation; no fairness audit; the one-half metals subsample in 2013-2014; the outcome is an epidemiologic classification; composites differ in content for participants with three scores; the 0.02 margin has no external justification	done
20	Interpretation, considering objectives and prior evidence	Discussion	done
21	Potential clinical use and implications	Decision-curve analysis (Fig. 4) shows no threshold at which the panel offers a material gain in net benefit	done

Note on why this checklist is worth filing. Of the seven comparison studies identified in the Background, none report to TRIPOD or TRIPOD+AI, and none report discrimination, calibration, cross-validation, or decision-curve analysis at all. Items A3 (leakage prevention), 13c (pre-registration) and 11 (performance measures) are where this study differs most sharply from that literature, and they are the items a prediction-model reviewer reads first.